

Part II

Injury Prevention

In part I we went back to school for an update on lumbar function. In part II this foundation is used to justify the best injury prevention approaches. The next two chapters discuss multiple models for risk assessment and ways to reduce the risk of low back injury.

Chapter 6

LBD Risk Assessment

Two types of physical risk factors predispose people to developing low back disorders (LBD): those linked to the person (e.g., movement competency and muscle endurance or the lack thereof) and those linked to the demands of performing a certain task (e.g., applied loads). This chapter describes how to assess the risk of back disorders that result from task demands.

Upon completion of this chapter, you will understand the variables that are important to include for risk assessment and be familiar with tools used to assess the risk. These tools are the NIOSH approach, the Snook psychophysical approach, the lumbar motion monitor (LMM) approach, and the Ergowatch approach.

Tissue overload causes damage and subsequent low back troubles. Assessing the risk involves comparing applied loads with some form of load-bearing tolerance value. Chapter 4 included examples of applied loads, tissue loads, and injury scenarios. Although direct measurement of tissue loads would be ideal, this remains impractical for large numbers of people performing a wide variety of occupational tasks and activities of daily living. Researchers have developed modeling approaches to predict these loads, but they remain methodologically complex. For this reason, surrogates for tissue load that are linked with posture, applied load, and motion have been proposed. These were introduced in chapter 2. The first section of this chapter, Brief Review of the Risk Factors for LBD, provides a list of known risk factors that includes descriptions and critiques of several approaches to assessing them.

Four risk assessment approaches are presented more or less from least complex to most complex. The simplest approaches use metrics that are the easiest and cheapest to employ for injury risk assessment. However, these are compromised by a lack of accuracy and specificity and are not sensitive to individual worker or person techniques. The more complex approaches involve more sophisticated methods and are more expensive to conduct, but they are more robust in their sensitivity to specific tasks and the unique ways

people elect to perform them.

Brief Review of the Risk Factors for LBD

Here are the risk factors for LBD that have been identified from epidemiological approaches:

- Static work postures, specifically prolonged trunk flexion and a twisted or laterally bent trunk posture
- Seated working postures
- Frequent torso motion, high spine rotational velocity, and spine rotational deviations
- Frequent lifting, pushing, and pulling
- Vibration exposure, particularly seated whole-body vibration
- Peak and cumulative low back shear force, compressive force, and extensor moment
- Slips and falls

The following risk factors have been identified from tissue-based studies:

- Repeated full lumbar flexion
- Time of day (or time after rising from bed)
- Excessive magnitude and repetition of compressive loads, shear loads, and torsional displacement and moments
- Insufficient loading so that tissue strength is compromised
- Rapid ballistic loading (such as that resulting during landing from a fall)

And finally, these personal variables have been identified as risk factors:

- Increased spine mobility (range of motion)
- Lower torso muscle endurance
- Perturbed or awkward motor control patterns
- Age
- Gender
- Abdominal, or torso, girth

NIOSH Approach to Risk Assessment

For about the past 40 years both field surveillance and laboratory studies have focused on the relationship between back injury and compressive forces experienced by the lumbar spine. For example, in 1981 the National Institute for Occupational Safety and Health (NIOSH) in the United States published guidelines for maximal compressive loading of the lumbar spine (both an action limit of 3,400 N and a maximal permissible limit of 6,300 N) based on some earlier evidence. In 1994 Leamon questioned the use of compression when he quite correctly stated that the supporting evidence for compression as a metric, or index of risk, was sparse. However, since that time, several good data sets have shown that restricting the amount of low back compression (both peak and cumulative) is one valid approach to reducing the risk of low back injury (e.g., data sets in the studies by Norman et al., 1998, and Marras et al., 1995). In addition, NIOSH has also proposed lifting guidelines to limit the amount of load lifted in the hands. These limits remain popular and are described in the following sections.

The influence of NIOSH has been far-reaching; many groups have used these values to reduce the risk in workers. In fact, the NIOSH approach continues to be widely used today primarily because of its ease of use.

1981 Guideline

The first lifting guide (NIOSH, 1981) was restricted to lifts in the sagittal plane and lifts that involve only slow and smooth motions. Predictions of the load lifted in the hands were based on some rudimentary distances to characterize the kinematics of the lift and load moment at the low back. Two limits were defined: the action limit (AL), which, if exceeded, triggered action to apply engineering and administrative controls, and the maximal permissible limit (MPL) for the load lifted, above which the risk is too high and not permissible. Weights lighter than the AL are considered safe. Those with biomechanical, physiological, and psychophysical expertise chose the magnitude of these limits and the variables needed to compute them. The general form of the formula used to compute the AL for weight in the hands is as follows:

$$AL \text{ (kg)} = 40 (HF)(VF)(DF)(FF)$$

where

- HF = horizontal factor, or the horizontal distance (H) from a point bisecting the ankles to the center of gravity of the load at the lift origin. This was defined as $(15 / H)$.
- VF = vertical factor, or the height (V) of the load at lift origin. This was defined as $(0.004) / (V - 75)$.
- DF = distance factor, or the vertical travel distance (D) of the load. This was defined as $0.7 + 7.5 / D$.
- FF = frequency factor, or the lifting rate defined as $1 - F / F_{max}$.
- F = average frequency of the lift, whereas F_{max} is obtained from tabulated data.

The logic of the equation is to set a maximal load of 40 kg (88 lb) and multiply this value against variables that act as discount factors. Thus, the maximal lift is 40 kg (88 lb) under optimal conditions and when all discount variables equal unity (1). Suboptimal lifting conditions cause the discount multipliers to drop to smaller values, reducing the safe load. The MPL is computed as three times the AL for a particular set of lifting circumstances:

$$MPL = 3(AL)$$

1993 Guideline

The revised NIOSH equation (Waters et al., 1993) was introduced to address those lifting tasks that violated the sagittally symmetrical lifting task restriction of the earlier equation. In addition, the concepts of the AL and MPL were replaced with a recommended weight limit (RWL) for a particular situation. If the actual load to be lifted exceeds the RWL, the risk of developing an LBD is elevated. Although the RWL equation is similar to the 1981 equation, two additional factors were incorporated: an asymmetrical variable for nonsagittal lifts and a factor for whether the object has handles. Note that the variable weightings also changed from the 1981 equation form.

$$RWL (kg) = 23(25 / H)[1 - (0.003 / V - 75)]$$

$$3 [0.82 + 4.5 / D])(FM)[1 - (0.0032A)](CM)$$

where

- H = horizontal location forward of the midpoint between the ankles at the origin of the lift. If significant control is required at the destination, then H should be measured both at the origin and destination of the lift.
- V = vertical location at the origin of the lift.
- D = vertical travel distance between the origin and destination of the lift.
- FM = frequency multiplier is obtained from a table supplied by NIOSH.
- A = angle between the midpoint of the ankles and the midpoint between the hands at the origin of the lift.
- CM = coupling multiplier ranked as good, fair, or poor. It is obtained from a table.

The 1993 equation predicts smaller loads that can be lifted safely, when compared to the 1981 equation, and is thus more conservative. Interestingly, NIOSH offered no provision for the difference in capacities of men and women; they are treated similarly largely for political reasons. (I discuss this issue of discrimination and the impact on protecting vulnerable workers in chapters 1 and 2.) Further, the approach ignored any individual differences in body mechanics used during lifting. In addition, some have suggested that the handle factors may not be consistent with the subsequent forces endured by the body. For example, although NIOSH assumed that having handles on the lifted object is better, our work has shown that handles allow the lifter to apply even more force, resulting in subsequently higher back loads (Honsa et al., 1998)!

Nonetheless, the equations form a rudimentary approach to risk assessment that is easy to conduct. Marras and colleagues (1999) concluded that the 1981 NIOSH guide identified low-risk jobs well (specificity of 91%) but did not predict the high-risk jobs well. The 1993 guide correctly identified 73% of the high-risk jobs but did not identify the low- and medium-risk jobs well. Overall, both NIOSH guides predicted jobs resulting in LBD with odds ratios between 3.1 and 4.6. Marras and colleagues (1999) noted that the most powerful variable was average low back moment (load in

hands and the horizontal distance between the anteriorly placed load from the spine).

Subsequent work from the Marras lab (Ferguson, Marras, and Burr, 2005) has added perspective to the interpretation of the NIOSH approach. The researchers measured populations of workers with low back pain and age-matched controls and noted that lifting from the floor is problematic for many workers, in particular those with back pain. This has implications for return-to-work programs for people with injured backs. Further, their data have strengthened the growing impression that shear forces on the spine are probably more important than compression in determining injury risk. In fact, 74% of their high-risk tasks were categorized based on excessive lumbar shear. Norman and colleagues (1998) made similar observations.

Potvin (2014) revisited the original criteria on which the revised equation was based and concluded that the equation is extremely conservative. For example, the RWL results in lumbar compressive forces that are far too low when lifting above knuckle height. Similarly, the equation RWLs are acceptable to 95% of females, which far exceeds the stated objectives when originally proposed. In summary, he found the RWL to be an average of 63% of the calculated composite acceptable loads based on the integration of the psychophysical and physiological criteria. Put another way, one would have to multiply the RWL by 1.65 to meet the original criteria put forth by NIOSH.

Snook Psychophysical Approach

The psychophysical approach to setting load, or work limits, is based on people's beliefs of what constitutes a tolerable work rate. The main criticism of this approach is that most people perceive physiologically related discomfort and not the internal loading on tissues that actually causes damage. For example, Dul, Douwes, and Smitt (1994) made a case that muscle discomfort, which is readily perceived, may not correlate with actual tissue loads. Workers may not be cognizant of the tissue loads as they reach damaging levels (McGill, 1997). Furthermore, Karwowski and colleagues (Karwowski, 1991, 1992; Karwowski and Pongpatanasuegsa, 1989) noted in a series of studies that the general psychophysical approach is very dependent on such factors as the instructions provided to the experimental subjects and the color of the object being lifted, to name a couple. For example, the Snook studies (1978) advised the workers to determine a work rate and load to be moved based on the instruction not to become unduly fatigued. The workers selected the loads they perceived not to be fatiguing. Yet when Karwowski repeated the experiments but changed the instruction to lift loads so that subjects would not become injured, the acceptable loads changed (they chose smaller loads). Karwowski and Pongpatanasuegsa (1989), when evaluating the effect of the color of the boxes being lifted, noted that workers would lift more when the boxes were white, because they perceived the black boxes as being denser. Recent work by my colleague Mardy Frazer (unpublished) suggests that females base their perceptions of task-limiting loads more on shoulder discomfort than on spinal indicators. Males, on the other hand, base their perception on low back limitations. The psychophysical approach appears to depend on many factors that modulate perception.

The work of Snook (Snook, 1978; Snook and Ciriello, 1991) is probably the most widely recognized in the psychophysics area. The investigators experimentally controlled variables such as object size, lift height, and movement distance in lifting and pushing and pulling tasks for both men and women. They constructed tables compiling the acceptable loads for both men and women over a variety of tasks. On one hand, this approach inherently incorporates the dynamics of the tasks, whereas on the other hand, the concerns raised in the previous paragraph are worth considering. Nonetheless, the Snook approach remains popular for its ease of

implementation and because it is one of the few assessment methods for pushing and pulling tasks.

Recent analysis has expanded the utility and interpretation of the Snook tables. Potvin (2012a) assessed the existing psychophysical data in a meta-analysis and noted that a strong, negative, exponential relationship exists between duty cycle and maximal acceptable effort (normalized to maximal voluntary effort). This means that the more time spent performing a repetitive task without rest greatly reduced perceived tolerance. No agreement exists on constant contractions with no rest. The equation originally developed for upper-extremity work, and subsequently confirmed for lifting and lowering tasks (hence the back; Potvin, 2012b), was as follows:

$$MAE = 1 - [DC - 1 / 28,800]^{0.24}$$

where

- MAE = average maximum acceptable effort, where 1.0 represents 100% MVE.
- DC = duty cycle, where 1.0 represents 100% of the cycle.
- 28,800 = number of seconds in 8 hours (its inverse is 0.0000347 and, when subtracted from DC, results in MAE = 1.0(100%) for all DC ≤ 1 s/workday).

The best fit exponent of 0.24 well predicts the MAE ($r^2 = 0.87$).

Lumbar Motion Monitor (LMM)

Several studies have documented the links between spine motion and the development of LBD, but none more thoroughly than those of Marras and colleagues (Marras, 2008; Marras et al., 1993, 1995). They developed several types of regression models that demonstrated that spine motion and, in particular, the velocity of motion and the range of motion are important predictors of those jobs with high rates of disorders (in fact, some odds ratios exceeded 10). The lumbar motion monitor is a three-dimensional goniometer that measures the three-dimensional kinematics of the lumbar spine (see [figure 6.1](#)). Marras and colleagues showed that the kinematic variables obtained from workers wearing the LMM, when combined with simple measurements of lift frequency or motion duty cycle and load moment (the load magnitude multiplied by the distance to the low back), provide impressive risk predictions. Using the LMM on people while they are performing real on-site tasks and not laboratory mock-ups is relatively easy to do. In addition, the LMM captures the unique ways people use and move their spines. See chapter 4 for a discussion of the mechanisms of injury that depend on spine posture and motion.

Figure 6.1 Wearing the lumbar motion monitor to record the three-dimensional kinematics of the lumbar spine during the performance of industrial tasks.



Courtesy of my good friend Professor Bill Marras.

Ergowatch

For approximately four decades, biomechanical models have been used to estimate loads in the low back tissues and identify high-risk jobs. Some models were intended as simple tools for health and safety personnel to use to approximate the index of injury risk on the plant floor. Other models were designed to be more robust in illustrating injury mechanisms while being sensitive to worker variance. Decisions as to which model to use boil down to the purpose and complexity of the model.

The better simple models need the complex models to assess the many simplifying assumptions that affect accuracy and output validity, which in turn depend on the type of application. Further, in many workplaces, the most blatant or overt ergonomic injury risks have been addressed, and only the more subtle risks remain. Ergonomists need simple models but also must be conversant with the more complex models that will assist in rectifying the subtler injury risks and in developing more effective intervention strategies.

Although simple biomechanically based models designed to obtain quick estimates of low back compression exist, risk assessment for most jobs requires more complex metrics and analysis approaches. To optimize biofidelity and industrial utility, Bob Norman, Mardy Frazer, Rich Wells, and I developed the software package Ergowatch. It uses the three-dimensional moments computed at joints while workers perform three-dimensional postures or tasks together with the fourth dimension of time and the effect of repetition on determining the safe load. (It is available by contacting Bbutler@uwaterloo.ca). The package uses the detailed output of the virtual spine (described in chapter 1) as a foundation, but makes assumptions to simplify data collection and facilitate routine analysis. In this way, individual worker behavior and spine mechanics are quantified but simplified into average muscular responses seen in workers performing similar tasks. This greatly simplifies data collection and preserves the benefits of better anatomical representation and more valid predictions of low back loads. Furthermore, by incorporating the injury data obtained from samples of workers, Ergowatch quantifies the risk of back injury during fully three-dimensional tasks and postures. The fourth dimension is time, in this case a variable needed to take into account the repetition of tasks over a work shift.

To use the Ergowatch, the user enters joint coordinate data manually or

manipulates an on-screen mannequin into the work posture of interest. The software executes the difficult task of determining the three-dimensional joint moments by computing the Euler angles and transforming them into moments about the orthopedic axes of each joint. In this way, the package is capable of calculating joint loads in any posture and for any combination of lift, lower, push, or pull task. It also incorporates algorithms that capture the average muscular response measured from workers to support three-dimensional spine moments of force together with strength data for both men and women. As well, the model is sensitive so spine posture in that ligament and passive tissue forces are accounted for during fully flexed spinal postures. Thus, the model accurately predicts low back shear forces supported by the spine (as compared to reaction shear forces predicted by other models), as well as compression forces that result from the load and the many torso muscles contracting to support a particular set of three-dimensional moments (see [figure 6.2](#), a and b) (note that shear more often is the variable closest to tolerance). Although the upper limit for shear has been suggested to be 1,000 N, the equivalent action limit for repeated loading has been set at 500 N (McGill et al., 1998, from tissue-based data; and Norman et al., 1998, using industrial injury reporting data). Finally, the risk is also calculated from accumulated loads during repetitive work via comparison of the load–time integrals with epidemiological data obtained from a large surveillance study conducted in an automotive assembly plant (Norman et al., 1998). (See [figure 6.3](#), a-d.)

Figure 6.2 Ergowatch computes lumbar compressive load from an algorithm representing the average response of the many muscles, measured from real workers, that combine to support the infinite combinations of three low back moments: flexion–extension, lateral bend, and axial twist. Because only three dimensions can be graphed at one time, (a) shows the surface of lumbar compression from combinations of flexion–extension and twist moments, and (b) shows the surface of compression from flexion–extension and lateral bend.

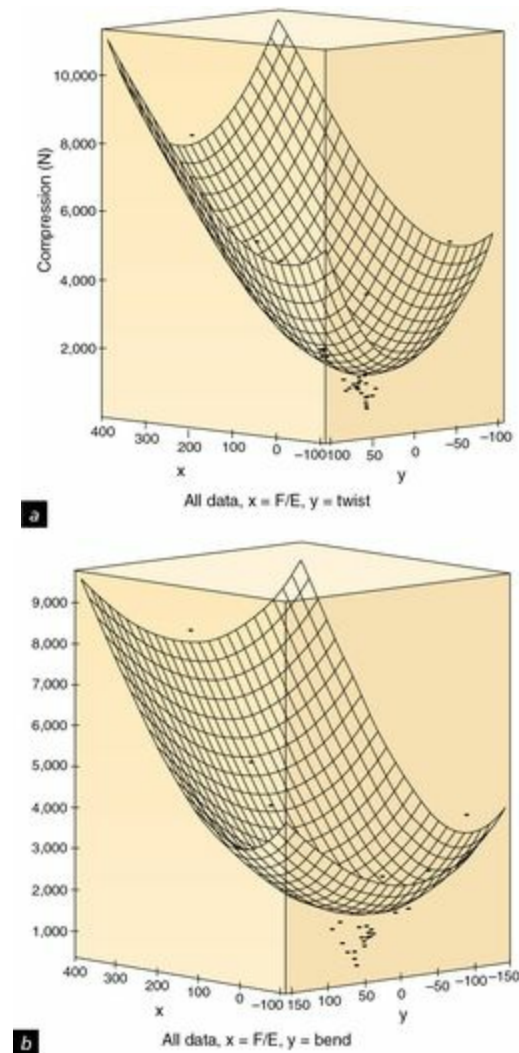
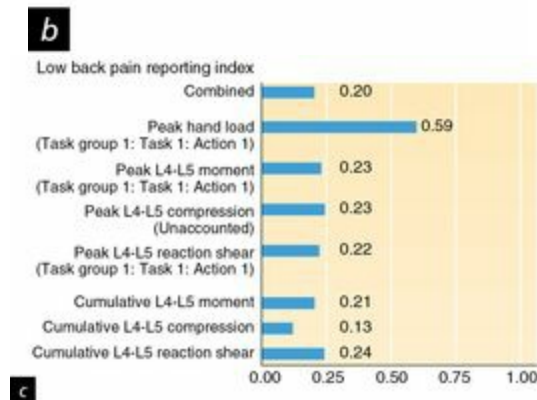
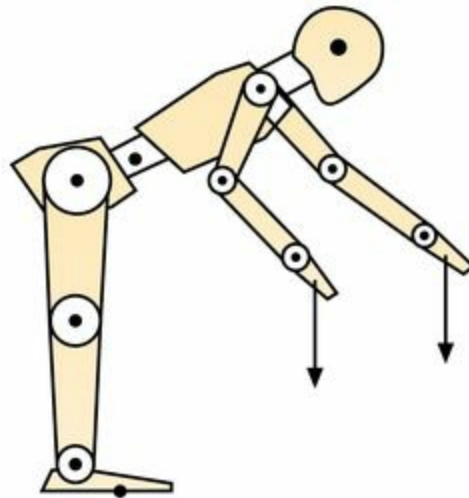
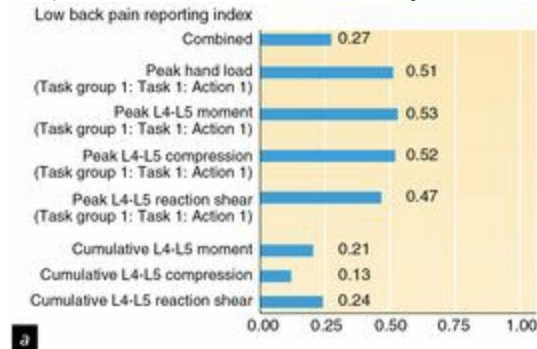
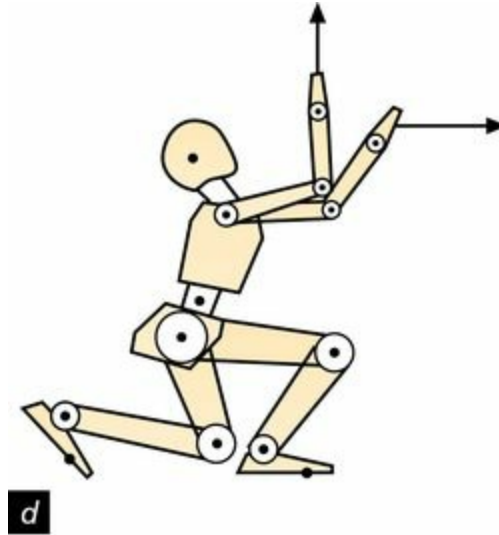


Figure 6.3 Ergowatch computes the joint-specific, three-dimensional orthopedic moments at each joint and uses several metrics to calculate risk.

For example, the low back risk includes compression and shear load threshold values along with accumulated loads during repetitive work by comparing the load–time integrals with epidemiological data obtained from a large surveillance study. Two examples are shown: (a, b) a reaching lift and (c, d) an overhead assembly task.





The Ergowatch approach represents a compromise between simple models that are easy to implement but are not very robust in their risk assessment and more complex approaches. Ergowatch is still relatively easy to use, and it provides some degree of sensitivity to the way individual workers perform complex tasks. It uses many risk indexes to quantify the subsequent risk from a specific task performed over a work shift.

Euler Angles and Orthopedic Moments

Estimations of three-dimensional joint moments are typically performed using three orthogonal axes (XYZ). But as a person moves, the orthopedic axes of the joints (e.g., in the lumbar spine, axis 1—flexion–extension, axis 2—lateral bend, axis 3—axial twist) also move so that they no longer align with the inertial XYZ axes. The difference between the orthopedic joint axes and the XYZ axes is described with Euler angles. The orthopedic moments are then obtained from the E-vector approach described by Grood and Suntay (1983). This involves taking the cross-product of the long axis (the primary axis, usually twist) of two adjacent segments to form the secondary axis (usually flexion–extension). The cross-product of the primary and secondary axes forms the E-vector, which in turn becomes the third joint orthopedic axis (usually lateral bend). Moments computed about axes XYZ can now be transformed into joint-specific orthopedic axes. This technique ensures that the joint axes coordinate system “stay with the joint” as the person moves within the inertial coordinate system.

Biological Signal–Driven Model Approaches

The final approach for risk assessment is to measure biological signals from each subject to capture the unique ways people perform their jobs and then use sophisticated anatomical, biomechanical, and physiological relationships to assign forces to the tissues.

Marras Model and McGill Model

The Marras model (Marras and Sommerich, 1991a, 1991b) measures electromyograms (EMGs) from several muscles and, using known physiological relationships, assigns forces to the muscles during virtually any industrial task. As noted earlier, this approach revealed powerful evidence linking the physical demands of specific occupational tasks with the incidence of LBD.

The McGill model (see chapter 1) uses the same philosophical approach to assign forces to the muscles, but attempts to include the highest level of anatomical accuracy possible. For example, it also measures three-dimensional spine curvature to assess forces to the passive tissues, including the intervertebral discs and the ligaments. By assigning forces to muscles and passive tissues throughout the full range of spine motion, it captures the ways people perform their jobs and even how they change with repetitions of the same jobs. The obvious liability of the approach is its enormous computational requirements, postprocessing of data, and difficulty collecting such comprehensive data from a wide number of workers in the field.

The question of the validity of this type of model must be addressed along with other models based on the biological approach. Some have argued that because these models contain known biomechanical and physiological relationships, they contain a certain amount of content validity. Moreover, both the Marras model and the McGill model have been quite successful in estimating the passive tissue and muscle forces that sum together to produce flexion and extension, lateral bend, and axial twisting moments. (These moments have been well predicted, with the exception of the twisting moment.) In summary, if twisting is not a dominant moment of force in a particular job, these models appear to predict accurate distributions of forces

among the support tissues.

EMG-Assisted Optimization Approach

Perhaps the current pinnacle of model evolution is a hybrid modeling approach known as EMG-assisted optimization (developed by Cholewicki and McGill, 1994) with stability analysis and tissue load prediction (Cholewicki and McGill, 1996, was an early example, and Ikeda and McGill, 2012, is a more recent example). This approach exploits the asset of the biological EMG approach to distribute loads among the tissues based on the biological behavior of the subject. It then uses optimization to fine-tune the balancing of joint torques about several low back joints. The optimization takes the biologically predicted forces and adjusts muscle forces the minimal amount possible to satisfy the three-dimensional moment axes at every joint over the length of the lumbar spine. Once the three-dimensional moments have been assigned and tissue forces determined, an analysis of spine stability is performed via comparison of the joules of work imposed on the spine through perturbation with the joules of potential energy existing in the stiffened column.

This most highly evolved of spine models provides the biomechanist or ergonomist with insight into injury mechanisms caused by instability (as witnessed by Cholewicki and McGill, 1992) such as occurs while picking up a very light load from the floor. Even though patients have reported back injury from incidents such as bending over to tie a shoe, previous modeling approaches were sensitive only to tissue damage and injury scenarios produced from large loads and moments. Now a biomechanical explanation is available to explain the subsequent tissue damage, and a method is available to detect the risk of its happening. Although the routine use of this type of sophisticated model by ergonomists is not feasible, it is useful (for trained scientists) for analyzing individual workers and identifying those who are at elevated risk of injury because of faulty personal motor patterns.

Simple or Complex Models?

In summary, the complex models provide a tool to investigate the mechanisms of injury and the effects of technique during material handling

on the risk of injury. The most complex and most highly evolved models provide insight into how injury occurs with all types of heavy and light loads. On the other hand, the simpler models, although sacrificing accuracy, can be powerful tools for the routine surveillance of physical demands in the workplace, but they must be wisely interpreted in each case given their limitations and constraints. Biomechanics and ergonomics require the full continuum of models. The choice of which one to use depends on the issue in question.

A Final Note

This chapter summarized and critiqued several approaches for their assets and liabilities. The choice of the most appropriate tool depends on the objective at hand. Yet many ergonomists and industrial engineers believe that minimal tissue loading is best for all jobs, motivating them to direct their interventions toward making all jobs easier. This concept is faulty. Biological tissues require repeated loading and stress to be healthy. Virtually all risk assessment tools consider only the risk of too much load. Risk assessment tools that survey too little loading for optimal health are just starting to emerge. The challenge is to develop a wise rest break strategy to facilitate optimal tissue adaptation. More robust assessment tools will be developed in the future based on the healthiest combination of work and rest. This will be achieved first through an understanding of the biomechanical, physiological, and psychological parameters of injury and human performance and then through thoughtful applications of this wisdom.